

Naveen Prabakar

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EDUCATION

Iowa State University

Ames, IA

Bachelor of Science in Computer Science, Minor in Data Science & Applied AI

May 2026

- **GPA: 3.87/4.00**, Dean's List (Jan 2023 – Present), President's List
- **Relevant Coursework:** Data Structures, Algorithms, DBMS, OOP, Computer Architecture, Discrete Mathematics, Software Development Practices, Data Science, Theory of Computation, Advanced Programming Techniques

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, JavaScript, TypeScript, Go, HTML, CSS

Databases: MySQL, MongoDB, Neo4j, AWS RDS, VectorDB, SQL Server, Cassandra, Redis, PostgreSQL

Tools: Git, Docker, Postman, Maven, Gradle, CI/CD, AWS (S3, EC2, ECR, Lambda, API Gateway, DynamoDB), OpenAI GPT, Knime

Frameworks: Flask, SpringBoot, React, Tailwind CSS, FastAPI, Express, Node, LangChain

WORK EXPERIENCE

Software Engineer (Senior Capstone Project)

01/26 – 05/26

Perpendo AI

Remote

- Collaborated with a team of 4 to build an MVP of an **agentic AI platform** to parse insurance underwriting documents and automate **risk evaluation** using LLM-driven reasoning.
- Developed full-stack features with **React, Python, and PostgreSQL**, supporting document ingestion and structured data extraction.
- Implemented automated **CI/CD pipelines** for services deployed across **AWS (EC2, DynamoDB, Lambda, API Gateway)**, enabling production updates and infrastructure changes to be deployed within **15 minutes**.

AI Software Engineering Intern

09/25 – 05/26

Merrill Manufacturing

Remote

- Engineered a full-stack Agentic AI assistant using React, Flask (Python), SuiteScript, and NetSuite's **LLM module** to let employees query **live ERP data** in natural language (products, customers, forecasts, maintenance).
- Built a **Retrieval-Augmented Generation** system over a product catalog with a **PostgreSQL conversational memory layer**, achieving end-to-end retrieval latency under **15 seconds**.
- Developed an **admin dashboard** tracking token usage, user activity, and security metrics, and implemented pandas-driven visualizations for sales and forecasting data.
- Created a **NetSuite Map/Reduce automation** to reconcile item receipts with carrier billing data, reducing manual reconciliation effort and improving workflow efficiency by **20%**.
- Implemented a **DeBERTa-based prompt-injection defense**, reducing malicious prompt success and improving response latency by **20%**.
- Developed an internal **MCP server** for Claude usage observability across **15–20 employees**, logging interaction metadata, request metrics, and estimated token usage.
- Provisioned and deployed the system on **AWS EC2**, configuring **EBS storage** and an **NGINX reverse proxy** for secure production hosting.

Database Administrator Intern

05/25 – 08/25

Uline

Pleasant Prairie, WI

- Optimized **2–3 SQL Server queries weekly**, achieving up to **42% reduction** in execution times, collaborating with the Senior DBA on performance tuning.
- Created **stored procedures** to detect log reader agent failures and partnered with the ESM team to integrate alerts into Nagios, improving replication visibility.
- Handled **20+ database operations** including permission management, backups and UAT restores, stored procedure deployments across SYS/UAT/PROD, SSRS uploads, and automated tasks using **VBScript**.

Lead Computer Science Tutor

08/23 – 05/25

Iowa State University Help Room

Ames, IA

- Led group tutoring sessions (4–5 students) across six CS courses, including **Data Structures, OOP in Java, Discrete Math, and Algorithms**.
- Mentored 100+ students in advanced algorithmic concepts and Java OOP principles over 18 months.

Crypto Analytics Platform | *Go, React, TypeScript, Cassandra, AWS*

06/25 – 09/25

- Architected a scalable microservice platform with **Go backend APIs**, **React/TypeScript frontend**, and **AstraDB Cassandra** for time-series data storage, deployed across 3 cloud providers (Vercel, Render, AWS EC2) with independent scaling capabilities, processing **10k+ records/week**.
- Engineered automated ETL pipeline with **Go cron jobs** ingesting crypto data every **10 minutes** from CoinGecko API, implementing **retry logic**, idempotent bulk inserts, and optimized Cassandra schema design achieving **sub-second query performance**.
- Built comprehensive analytics system generating **daily automated PDF reports** with email distribution, featuring real-time dashboards maintaining **99.9% uptime** for continuous data processing and visualization.

School Subject Assistant | *Python, Redis, Pinecone, JavaScript, HTML, CSS, AWS, Docker*

08/25 – 09/25

- Built full-stack AI Q&A system for statistics using **1,000 PDF pages**, **Pinecone vector search**, and **OpenAI embeddings** for context-aware solutions.
- Deployed **serverless architecture (Docker → AWS ECR → AWS Lambda → API Gateway)** with **Redis session memory**, achieving minimal downtime; approximately 10 seconds per cold start

AI-Powered Resume Q&A System | *FastAPI, OpenAI GPT, Python, MongoDB, AWS S3, FAISS*

06/25

- Built an intelligent **Q&A system** that answers questions about PDF resumes using OpenAI's GPT via FastAPI.
- Implemented **PDF chunking**, **vector embedding (FAISS)**, and **S3-based self-learning** pipeline.
- Integrated **MongoDB log management system** to track user interactions and improve model responses.
- Enabled real-time document understanding and adaptive learning across user sessions.

MCP-Auto-ML | *Python, Pandas, Scikit-learn, AWS S3, MongoDB, Claude AI*

04/25 – 05/25

- Engineered AutoML system based on **Model Context Protocol**, enabling **direct code execution and deployment**.
- Created end-to-end ML pipeline: **data ingestion**, **cleaning**, **feature encoding**, and **model selection**.
- Achieved **94.57% accuracy** on heart disease prediction task using optimized ML models.
- Integrated **AWS S3 for model storage** and **MongoDB for data persistence and monitoring**.

E-commerce Phone Store | *React, Tailwind CSS, Node.js, Express.js, MongoDB*

03/25 – 05/25

- Built full-stack online phone store with **product listing**, **cart management**, and **checkout flow**.
- Implemented responsive design and client-side form validation for improved UX.

RogueLike Dungeon Crawler | *C, C++, Linux, Ncurses*

01/25 – 04/25

- Engineered a feature-rich roguelike game in C/C++ using **ncurses**, with **procedural level generation**, turn-based combat, and modular game architecture supporting dynamic difficulty scaling.
- Designed and optimized core systems including **enemy AI pathfinding Dijkstra's Algorithm**, and a full-featured **fog-of-war + line-of-sight rendering pipeline**.

AI-Powered Note-Taking App | *Java, Spring Boot, MySQL, WebSockets, CI/CD*

08/24 – 12/24

- Designed backend infrastructure using **Spring Boot** and **MySQL**, optimizing data retrieval by 20%.
- Integrated OCR functionality with **Tesseract API**, achieving 90% text extraction accuracy.
- Developed real-time live editing features via WebSockets, reducing collaborative latency by 30%.

Discord Anime Trivia Bot | *Python, MySQL, Docker, AWS (EC2, S3, RDS)*

05/25 – 06/25

- Developed multilingual anime trivia bot supporting **multiple servers with real-time score tracking** and persistent MySQL storage.
- Deployed on AWS EC2 with MySQL database hosted on RDS and quiz images stored in S3, ensuring **scalability, reliable infrastructure, and minimal downtime**.

Coding Journal | *Python, MySQL*

09/24 – 12/24

- Developed a terminal-based journaling application in **Python** with full **MySQL database integration** to store, retrieve, and manage personal journal entries.
- Implemented functionality for **creating, editing, and querying** journal entries by date, enabling efficient information retrieval through SQL queries.

Machine Code Translator | *Java, LEV8 Assembly, Bit Manipulation*

05/24 – 08/24

- Built a Java-based system to decode 32-bit machine code into **LEV8 assembly instructions** using low-level **bit manipulation techniques**.

- Implemented a parser to read binary files and translate them into human-readable assembly code, including support for **label resolution and conditional branching**.
- Achieved over **95% decoding accuracy** across instruction sets, verified through extensive testing and edge-case validation.

Neo4j & MySQL Database Management | *Cypher, SQL, ER Modeling*

01/24 – 04/24

- Designed and implemented relational and graph databases using **MySQL** and **Neo4j**, focusing on data integrity, normalization, and query performance.
- Performed extensive **DDL and DML operations** to define schemas, populate tables, and manage data across both SQL and graph-based environments.
- Imported and structured data from **CSV files** into MySQL and Neo4j, and optimized query logic using **SQL** and **Cypher** for efficient data retrieval.

LEADERSHIP & INVOLVEMENT

BackEnd Lead

Jan. 2025 – Present

AI/ML Club

Iowa State University, Ames, IA

- Conducted workshops showcasing Gemini API, focusing on fine-tuning models, prompt design, and multi-modal (image) prompts.
- Taught the 5-step process of how Generative AI works, covering preprocessing, pre-trained models, fine-tuning, advanced fine-tuning, and model deployment.
- Show cases capabilities of Model Context Protocol through workshops by making a graphing tool
- Taught Retrieval Augmented Generation through workshops based off reference to documents and images

CERTIFICATIONS

- **Prompt Design in Vertex AI**

Google Cloud, March 2025

Skills: Prompt engineering, multimodal generation, image analysis, Gemini model usage, generative AI in marketing.

- **AI Fundamentals**

DataCamp, June 2025

Skills: AI subdomains (Machine Learning, Deep Learning), Generative AI concepts, ethical AI, prompt construction.